

BS – 101 Engineering Physics

Lecture – 1

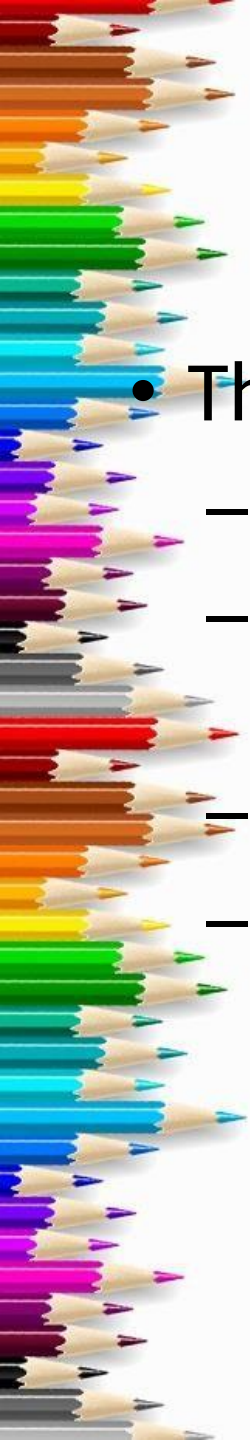
Engr. Safee Ullah





Subject

- Lectures
- Email: Safee.ullah@hitecuni.edu.pk
- Office No. 3 (Department of Electrical Engineering)
- Monday – Friday (Between University Timings)
- 75% Attendance is compulsory
- All course material is uploaded on google Classroom



Assessment Distributions

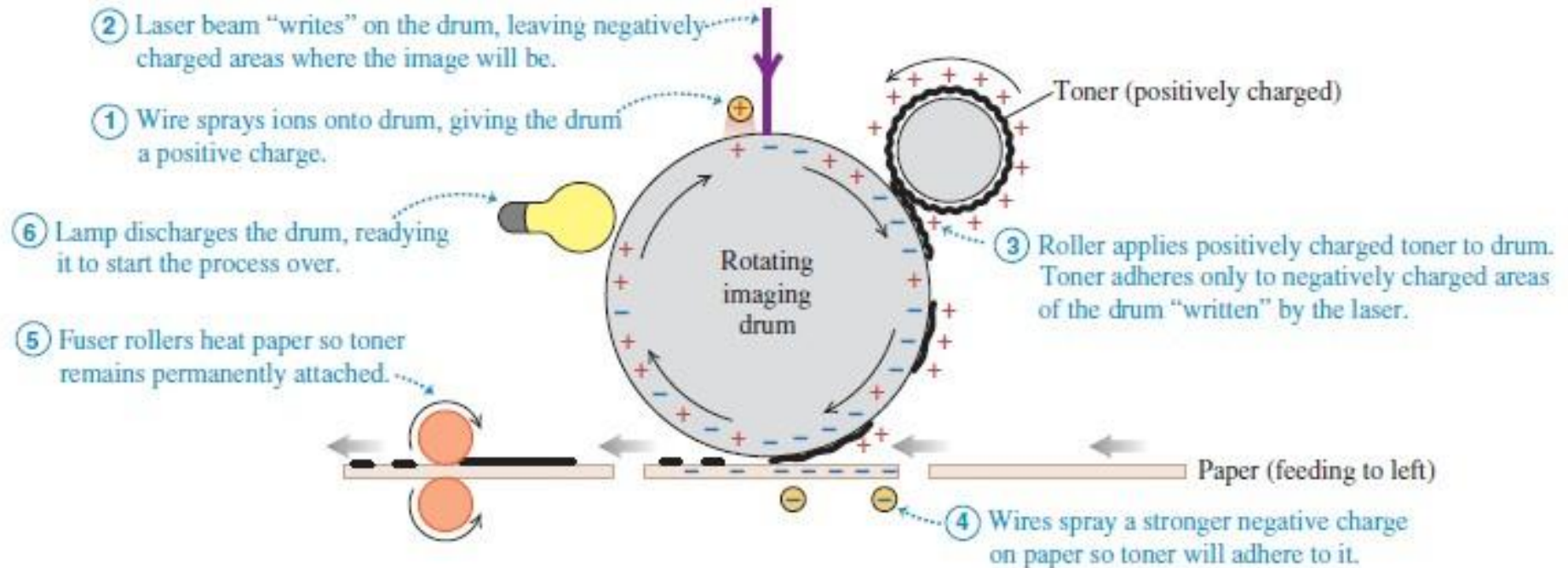
- Theory (100%)
 - Final (50%)
 - Sessional/ Mid (30% --- 15% each)
 - 6 Quizzes (10%)
 - 3 Assignments (10%)



Electric Charge

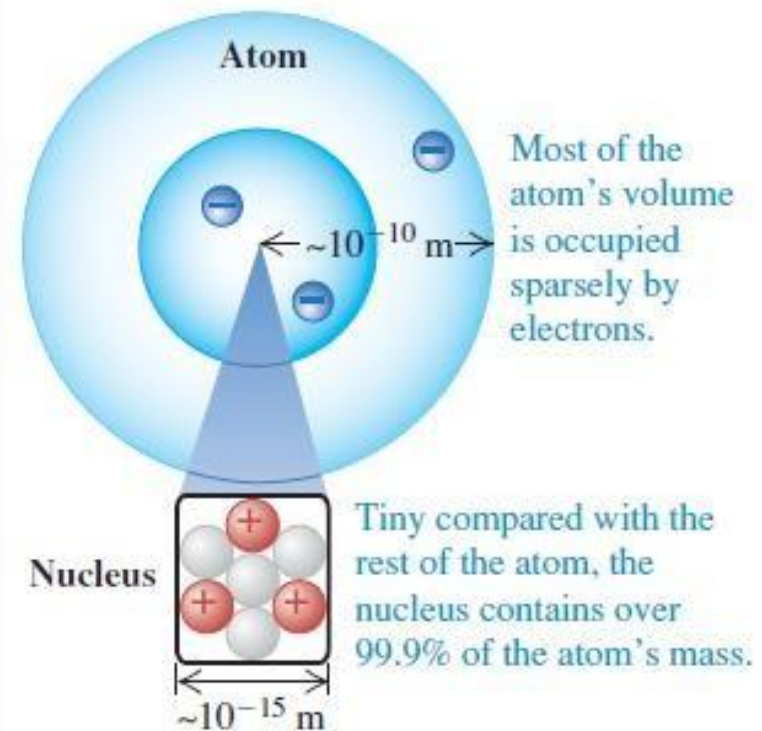
- Ancient Greeks discovered the phenomena of electric charge around 600 B.C.
- The word “Electric” is derived from the Greek word “Elektron” meaning amber.
- Like charges repel each other and unlike charges attract each other.
- Two positive or two negative charges repel each other
- A positive and a negative charge attract each other.
- It doesn't mean that their magnitudes must always be equal.

Electric Charge (Application)



Atomic Model

- Most of the atom's mass is concentrated in the nucleus. (about 99%).
- Electrons are lighter than the protons.
- They are easily moveable.



	Proton: Positive charge Mass = $1.673 \times 10^{-27} \text{ kg}$
	Neutron: No charge Mass = $1.675 \times 10^{-27} \text{ kg}$
	Electron: Negative charge Mass = $9.109 \times 10^{-31} \text{ kg}$

The charges of the electron and proton are equal in magnitude.



Materials

- Insulators
- Conductors
- Semiconductors



Coulomb's Law

- Charles Augustin de Coulomb (1736 – 1806)
- The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.



Coulomb's Law (Mathematical Form)

- $F = k \frac{q_1 q_2}{r^2}$
- Force (Newtons)
- Charges (Coulomb)
- Distance (meters)
- If the charges are same then the forces will be repulsive.
- If the charges are opposite then the forces will be attractive.



Constant of Proportionality

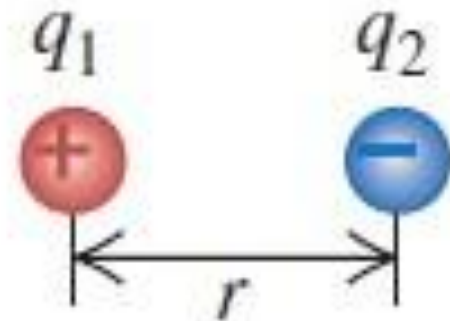
- k in Coulomb's law is the constant of proportionality. Its units are Newton meter square per coulomb square.

- $k = 8.988 \times 10^9 \text{ Nm}^2/\text{C}^2$

- $k = \frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ Nm}^2/\text{C}^2$

Coulomb's Law (Example)

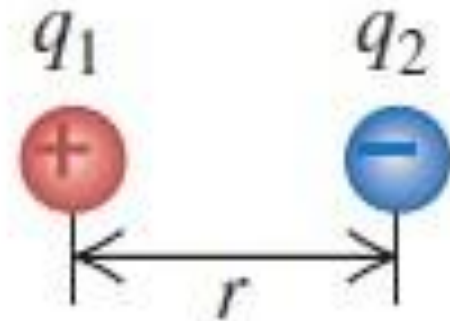
- Two point charges $q_1 = +25nC$ and $q_2 = -75nC$, and are separated by a distance $r = 3.0cm$. Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .



Coulomb's Law (Example)

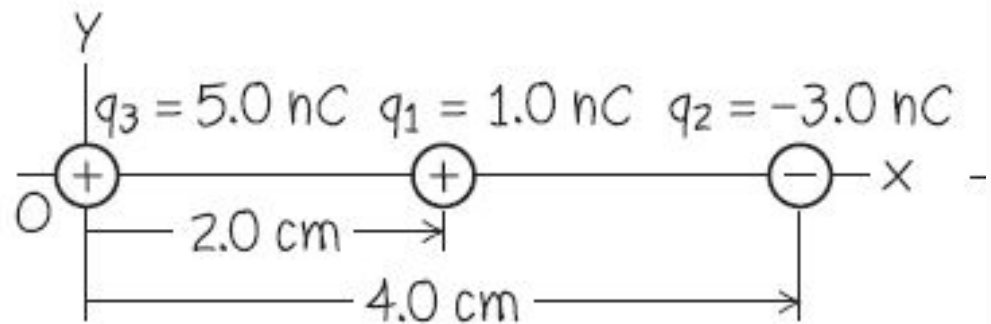
- Two point charges $q_1 = +25\text{nC}$ and $q_2 = -75\text{nC}$, and are separated by a distance $r = 3.0\text{cm}$. Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .

- 0.019 N



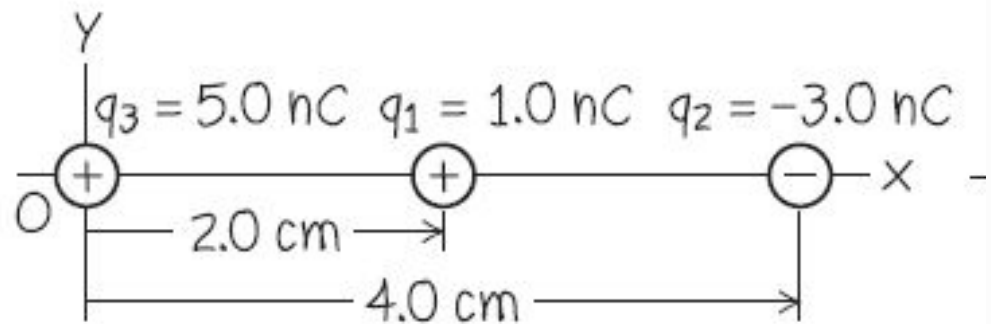
Coulomb's Law (Example Contd.)

- Two point charges are located on the x-axis of a coordinate system: $q_1 = 1\text{ nC}$ is at $x = +2\text{ cm}$, and $q_2 = -3\text{ nC}$ is at $x = +4\text{ cm}$. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5\text{ nC}$ at $x = 0$?



Coulomb's Law (Example Contd.)

- Two point charges are located on the x -axis of a coordinatesystem: $q_1 = 1\text{ nC}$ is at $x = +2\text{ cm}$, and $q_2 = -3\text{ nC}$ is at $x = +4\text{ cm}$. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5\text{ nC}$ at $x = 0$?



- $-28\mu\text{N}$